

Research paper

Co-optimization of power and water distribution systems for simultaneous hydropower generation and water pressure regulation

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ABSTRACT

Water Distribution Systems (WDSs) are gravitational systems supplying water, mainly. Little research has been conducted on harnessing the great amount of energy that is currently dissipated during water pressure regulation, usually performed with “passive” devices like Pressure Reduction Valves (PRVs). This paper develops a co-optimization framework between Water and Power Distribution Systems (PDS) and introduces the concept of Integrated Power-Water Distribution System (IPWDS). Under this framework, PRVs are replaced by hydro-turbines, which form the interconnecting links between the two systems. They regulate pressure at the WDS while injecting the power resulting from their action to PDS. The complete WDS and PDS mathematical models are adopted and the detailed turbine design and performance model that accounts for turbines feasible operational range is considered. The suggested method optimizes the power generation of the hydro-turbines subject to the hydraulic restrictions and power flow constraints imposed by each network. Optimal Power Flow (OPF) has been extended in order to accommodate turbine’s model and constraints imposed by the varying daily flows of the WDS and PDS. Results taken under the suggested framework from a typical WDS, demonstrated that more than 393 MWh (approximately \$40 K) of clean power could be produced per year. However, the violation of the unified set of constraints from the IPWDS may reduce the amount of renewable power generation by a significant 7%. This paper focuses mainly on the technical aspect of such an approach. Future work will focus on the economic assessment and viability of such joint projects, resulting from co-optimization and on social-technical aspects that aim on maximizing the penetration of turbines within existing PDS.

1. Introduction

Water and electricity are considered as the most critical resources for mankind. However, water and energy systems have been traditionally owned and controlled by separate utilities, hence they are operated interdependently from one another. Recent reports suggest that the interdependency between energy and water networks is set to intensify in the coming years (“WEO-”, 2016) and thus a joint operation between the two has become an emerging topic of vital importance. Moreover, the grand-challenge of implementing sustainable resource management and the intensification of water and energy consumption, further stimulate academia to conduct extensive studies for obtaining a comprehensive insight of their intrinsic linkage considering multiple criteria. Additionally, the ever-developing concept of energy community microgrids that satisfy their local load from distributed Renewable Energy Source (RES) has led researchers to direct their attention to evaluating all the potential local generating resources.

Studies that involve both systems are limited, and hence innovative approaches that improve their operation in terms of energy efficiency by means of coordinated operation, should be further investigated. Technical literature highlights the growing need of intersectoral co-operation for accomplishing circular economy and improving system sustainability (Bačević and Østergaard, 2018; Dall’Anese et al., 2017). An analytical review of the major water-energy studies and their key findings under different research methods and objectives has been conducted in (Hamiche et al., 2016). According to the available data drawn from water utilities and numerous research studies (Ferrarese and Malavasi, 2020), the average dissipated power through PRVs, is estimated around 10 kW and can reach up to 100 kW. It has also been calculated from the same data that if PRVs that dissipate more than 10 KW are replaced with energy recovery machines, then almost 70% of the total power dissipated within those passive devices would be recovered, while their payback period would range between 5 and 10 years. This approach may have a positive impact towards the sustainable operation

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and development of the system, specifically within an urban environment.

Several models for the inextricably connected power and water sectors have been developed that provide useful insights. References (Oikonomou and Parvania, 2020a; Zamzam et al., 2019) have developed a framework under which the operation of the controllable assets of the WDS supplied by the PDS, is optimized in terms of achieving energy cost minimization for both systems. A coordinated operation of pumped storage hydropower units that minimizes the operational costs of PDS is presented in Majidi et al. (2022), while a Mixed Integer Non-Linear Programming (MINLP) model for the maximization of energy savings in water-energy microgrid is formulated in Moazeni and Khazaei (2020). Additionally, Li et al. (2019) investigates the potential of water networks to provide demand response services to the PDS. All the aforementioned papers exploit the flexibility of the already installed and controllable assets of the water system: for example energy-consuming pumps used for water treatment and desalination purposes. They do not consider the significant, unexploited hydropower potential of the WDS existing at nodes of pressure regulation. Furthermore, even if existing bibliography examines simplistic water distribution test systems (Oikonomou and Parvania, 2020a; Zamzam et al., 2019; Majidi et al., 2022; Moazeni and Khazaei, 2020; Li et al., 2019; Oikonomou and Parvania, 2020b), it offers limited practicality, as they consist of a small number of nodes with no loops. They often simplify the problem further by simulating either pumps or Pumps as Turbines (PATs) as static devices, assuming simple head losses of constant speed (Li et al., 2019) or constant efficiency (Oikonomou and Parvania, 2020a; Zamzam et al., 2019; Majidi et al., 2022; Moazeni and Khazaei, 2020), (Li et al., 2019; Fecarotta and McNabola, 2017). Moreover, (Zamzam et al., 2019) ignores the line flow limits of the PDS, while (Fecarotta and McNabola, 2017; Corcoran et al., 2016; Giugni et al., 2014) focus on optimization of energy recovery within WDSs, addressing this issue only by the hydraulic network perspective, ignoring the impact energy has when injected to the PDS. While the employment of a hydro-turbine via a pumped-hydro storage scheme is introduced in Majidi et al. (2022), the proposed turbine offers no pressure regulation and its primary source of power cannot be considered as a RES, since the same water used for power generation has been previously pumped into the water tank by consuming power from the PDS. Additionally, since the placement of the hydro-turbine is restricted only to locations near a water tank, their operation does not guarantee optimal energy production as the highest energy potential may appear at another point of the water network.

In the WDS side, none of the aforementioned work addresses to the turbine's type, design parameters and actual dimensions, while assuming constant efficiency to both pumps and hydro-turbines. In the PDS side, they all model electric loads as constant P-Q sinks. This oversimplified model is inappropriate for PDSs with great variations of the voltage pattern throughout the day. Hydropower injection also affects the voltage profile decreasing the accuracy of the PDS demand model further. Inaccuracies in modeling of the aggregated steady-state behavior have a significant impact on the accurate simulation and optimization of the power grid steady-state response (Jereminov et al., 2019).

To the best of authors' knowledge, there is currently no published work that investigates the potential of implementing pressure regulating hydro-turbines in a WDS and their integration into a co-optimization problem that accounts both water and power systems constraints. Here, a mathematical turbine model is introduced in the optimization problem that involves the feasible operation range of a pressure regulating turbine of variable speed and efficiency and its interaction as generator with the PDS side. In this work, for the first time, turbine's type, design and dimensioning are incorporated into the optimization process, as it they are specifically determined to offer precise pressure regulation with maximized power generation efficiency. Furthermore, simulations have been conducted in a real water network of considerable scale. In this regard, the real impact of the turbine installation

through the whole network is assessed, so the final results are representative of a real case scenario.

The performance characteristic curves of turbines working as PRVs in fluctuating hydraulic operating conditions are addressed in Bideris-Davos and Vovos (2023). By considering the hydro-turbine's performance, along with the technical characteristics and restrictions of the WDS components, a more accurate and realistic coordination schedule between the interconnected systems can be achieved. The proposed mathematical formulation integrates the performance of the hydro-turbines for the varying WDS conditions into the classic OPF formulation, so that their dispatch does not violate neither WDS nor PDS constraints. As the urban hydro-generation is a form of RES, the PDS operators are usually obliged to absorb as much power from them as it is technically feasible by their system. In this regard, the objective function aims at maximizing their output while achieving the same pressure drop as the existing PRVs, while respecting PDS and WDS constraints and exploiting the otherwise dissipated energy. The rapid adaptability of the turbine rotational speed N in response to the water demand fluctuations, provides flexibility in both systems as pressure regulation and power generation are adjusted in such manner that hydraulic and electrical constraints are simultaneously fulfilled.

Specifically, the classic formulation of the AC OPF model has been expanded, so that it implicitly accommodates the hydraulic restrictions of the water network (pressure limits) and turbine operation (affinity laws), as additional constraints. Algorithms based on the Interior Point Method are widely applicable in solving OPF problems of large scale power systems with great number of variables and constraints as they are capable of reaching the global optimum solution, while displaying fast convergence rates. The nonlinear WDS balance equations and constraints introduced by the water network are enforced by calling and operating the "industry-standard" hydraulic simulator EPANET (Rossman, 2000), within the MATLAB environment. The EPANET solver is based on the Global Gradient algorithm, which is basically a variant of the Newton-Raphson method and can handle extended period hydraulic analysis of a WDS. The WDS balance equations include the mass continuity equations within the nodes and momentum balance equations along the water pipes and the headloss imposed by water circulation within these pipes. Due to the detailed turbine modeling, additional hydraulic constraints are introduced, stemming from the underlying physics that govern hydro-turbine's design and performance modeling. The turbine modeling within EPANET does not require special handling, as it is handled in a similar manner as a typical PRV, hence, no further complexity is added in the hydraulic simulation of the WDS. However, the integration of hydro-turbines on large scale systems increases the computational burden and creates new technical issues. When trying to integrate a large number of hydro-turbines into a co-optimization problem that involves a power flow analysis, the overall computational complexity is expected to intensify, as the operation of pressure regulating turbines introduces a set of additional non-linear equations to the problem. Nonetheless, employing a solver that can integrate parallel processing techniques will significantly reduce the computational effort. Regarding the technical issues, if turbines' dimensioning is not optimized so that their total size becomes comparable with PRVs' size, then a one-by-one replacement might not be practically feasible.

The main contributions of this work are summarized below:

- 1) It introduces an integrated methodological framework that holistically addresses the water-energy nexus by proposing hydro-turbines as interlinks between water and power systems at distribution level.
- 2) The optimization problem involves the complete water and power distribution system equations along with the detailed turbine performance model, while including a trade-off cost for WDS and PDS constraint violation.
- 3) The optimization problem includes the design and dimensioning process of hydro-turbines that offer water pressure regulation with maximized power generation efficiency. Therefore, turbine's design

and size are optimized according to the operating conditions and hydraulic properties of the location they are installed within the WDS.

- 4) The creation of a unified system model that can be easily extended to achieve an overall multi-objective optimization under different criteria.

The research stages of the work presented here started with the implementation of an algorithm that precisely replicates the operation of existing PRVs with the use of hydro-turbines in terms of achieving the same downstream pressure output. Then, a process that selects the turbine design that ensures the lowest pressure deviation from the PRV's ideal setpoint from all the feasible turbine designs at a specific location, has been developed. Furthermore, the produced turbine geometry is designed, so that it is always comparable in size with the replaced PRV so that this replacement is feasible in practice. Then, the OPF formulation has been expanded so that it can accommodate the power generation from all those locations, along with the various constraints introduced by their operation to both networks. The result is the optimal performance of both networks, according to a unified objective function that offers maximum possible hydrogenation, along with accurate pressure management. The framework is implemented on a full-scale network and not on simplistic water distribution test systems that are employed in related works mentioned above. Results taken from a typical IPWDS indicate the significant benefits that emerge from the co-optimization of the WDS and the PDS operation. Specifically, for the WDS side, it is demonstrated that turbines are capable of regulating water pressure as efficiently as PRVs, while harnessing the otherwise dissipated hydropower. For the PDS side, it creates a significant annual revenue from the energy injection, while minimizing the total generation cost of the system.

The remainder of this paper is organized as follows. In the second section, the WDS modeling is presented. The rationality behind PRV replacement with hydro-turbines is also explained. Section 3 describes the PDS model and the integration of hydro-generation within the classic OPF formulation. Section 4 defines the IPWPS and formulates the co-optimization problem, outlining the interaction between the inter-system constraints. Section 5 presents the PDS and WDS under study before the turbine integration and demonstrates the results derived from the proposed intersectoral optimization framework. Finally, Section 6 outlines the benefits of co-optimization, as indicated from the results and suggests future work and expansions of the suggested framework.

2. Water distribution system model

In this Section, we will present the rationality behind PRV replacement with hydro-turbines and the fundamental equations that govern the WDS and hydro-turbine operation. The intersystem constraints that are later incorporated into the final co-optimization problem, are also presented.

2.1. Water pressure regulation using hydro-turbines

When investigating the potential of employing energy recovery devices within a WDS, one of the biggest challenges is to locate their optimal location within the network. The approaches presented within the literature (Fecarotta and McNabola, 2017; Giugni et al., 2014; Jereminov et al., 2019) usually oversimplify the problem by means of ignoring the turbine's detailed model and selecting a priori a specific turbine type, without assessing the potential types with overlapping operational regions. Moreover, they tend to disregard the real feasible operating region of the turbine or estimating it roughly, by assuming a constant efficiency regardless of the water flow and head fluctuations throughout the day.

Regardless of the approach, the practical challenge of incorporating turbine operation within a WDS is the possibility of reverse water flows

within the pipes, that ultimately translate to discontinuous power production. More specifically, in water systems with dense looping characteristics, the flow within a pipe throughout the day changes direction. Unfortunately, flow through turbines cannot be reversed in such a way that they still generate power or offer pressure regulation (Fecarotta and McNabola, 2017). Therefore, it is vital to ensure a constant direction of flow at these locations. However, the exact impact of turbines performance in the whole network cannot be determined a-priori, and thus the identification of suitable pipes where a unidirectional flow is always maintained, is not possible before a multi-period hydraulic flow analysis of the WDS is performed. When the flow does not comply with turbine operation, the turbines are inactive and act as a by-pass for water discharge. When such case exists, turbine power production is characterized by discrete power generation levels which further intensifies the problem complexity.

On the contrary, discontinuous operation of hydro-turbines is not expected when they are considered as replacements for PRVs. Then, not only the obstacle of identifying the best locations for turbine placement is surpassed (Fecarotta and McNabola, 2017; Giugni et al., 2014; Jereminov et al., 2019), since excess pressure and high flow are expected, but also their operation is ensured to be continuous, because water flow is guaranteed to be unidirectional within the respective pipes. Finally, with this approach the best turbine type, its real feasible operating region and impact can be determined a priori, because the PRVs size and pressure requirements are specific and pre-defined.

According to Laghari et al. (2013), mini and micro hydropower schemes display the shortest payback period over all other small-scale RES technologies. For WDS locations with significant hydropower potential that appears in large pipes, their cost becomes comparable with the one of PRVs, based on the total PRV cost that is demonstrated in García et al. (2019) and by considering the lost profit due to their inability to harness the dissipated energy (Bideris-Davos and Vovos, 2023). Especially the one-by-one replacement of PRVs with turbines, employs existing network infrastructure and space and hence a substantial part of the civil work cost has already been paid. Therefore, the reduced capital cost compensates for the high initial purchase cost, providing even greater short-term benefits.

2.2. The hydro-turbine as link between WDS and PDS

The hydro-turbine constitutes the physical link between the two systems and their common coupling, i.e., the point wherein the power flows from WDS to the PDS. Turbine's real (P_T) and reactive (Q_T) power production represents the shared variable between the two systems. It encapsulates the optimization variables in a single formula by including the power generation (PDS side) and the flow rate (Q) and head (H) regulation (WDS side). The set of equations that govern the hydro-turbine model during the WDS hydraulic simulation, is explained in detail in Bideris-Davos and Vovos (2023), but they are also briefly described below for the completeness of the model presented here.

At all times, the water head values at each junction must be within a specific range that is pre-assigned by the local water system operator to meet the service quality pressure standards and prevent supply disruption incidents from components failure. These Minimum and Maximum Head Standards ($MHS_{\min}$ and $MHS_{\max}$, respectively), restrict hydro-turbine performance, forming additional head constraints assigned to each WDS junction. Therefore, hydro-turbines are allowed to impose head drops that result in junction head values that are kept within these statutory limits:

$$MHS_{\min}(m) \leq H_j(m) \leq MHS_{\max}(m), \forall j \in [1..n_j] \quad (1)$$

where H_j is the head value of junction j and n_j is the total number of WDS junctions.

The most commonly used empirical formula for calculating the pipe headloss H_p (m) is given by the Hazen-Williams (H-W) equation

(Swamee and Sharma, 2008):

$$H_P = A Q^B \quad (2)$$

where Q is the water flow rate (m^3/s) and A, B is the resistance coefficient and flow exponent of the H-W formula. Coefficient A depends on the pipe material, diameter, length and age and is approximated with the use of available table charts (Swamee and Sharma, 2008), while B is constant and equal to 1.852.

The head used by the turbine is calculated by:

$$H_T = f(Q) = \zeta_f \cdot Q^2 + \frac{u_2}{g} \left(\frac{1}{\pi D_2 b_1} \cdot \frac{1}{\tan \alpha_1} + \frac{1}{\pi D_2 b_2} \cdot \frac{1}{\tan \beta_2} \right) \cdot Q - \frac{1}{g} \left(u_2^2 + \frac{c_a}{2} + \frac{k \cdot w_c^2}{2} \right) \quad (3)$$

where ζ_f is the friction losses coefficient (dimensionless), g is the gravitational acceleration (m/s^2), u_2 is the peripheral blade velocity at turbine outlet velocity triangles (m/s), D_2 is the outlet runner diameter (m), b_1, b_2 is the blade width at turbine inlet and outlet blade, respectively (m), α_1 is the guide vane's pre-swirl angle (in degrees), β_2 is the blade outlet angle (in degrees), c_a is the water velocity at turbine outlet (m/s) and w_c is a velocity component (m/s) that indicates the shock losses on the runner blade with constant k ranging from 0.5 to 0.9.

The turbine's power output formula is:

$$P_T = \gamma \cdot Q \cdot H_T \cdot n_h \quad (4)$$

where γ is the specific weight of water ($9810 N/m^3$) and n_h is the hydraulic efficiency of the turbine.

Net power injection at each node i , is given by:

$$P_{net,i} = P_{G,i} + P_{T,i} - P_{L,i} \quad (5.a)$$

$$Q_{net,i} = Q_{G,i} + Q_{T,i} - Q_{L,i} \quad (5.b)$$

where $P_{G,i}$, $Q_{G,i}$ are the conventional generator real and reactive power outputs, $P_{T,i}$, $Q_{T,i}$ are the hydro-turbine real and reactive power outputs and $P_{L,i}$, $Q_{L,i}$ are the real and reactive power of the load at node i . Of course, any of those couples of variables are set to zero, if the respective element does not exist (e.g. no conventional generator at bus 3, then $P_{G,3} = Q_{G,3} = 0$).

A step-by-step methodology for turbine dimensioning and operation, so that they precisely mimic PRV operation is addressed in Bideris-Davos and Vovos (2023). According to the turbines' design and performance curves, their operation can be adjusted in such way that, for a specific Q in time the requested turbine H_T can be accomplished under a specific Pressure Regulation Deviation (PRD) value. PRD is determined by the ability of the turbine to replicate the PRV operation in terms of headloss at its outlet. It is expressed as the absolute percentage difference between the ideal (H_{PRV}) and turbine (H_T) headloss:

$$PRD \stackrel{\text{def}}{=} \frac{H_{PRV} - H_T}{H_T} (\%) \quad (6.a)$$

According to affinity laws, between different working points the following set of equations is satisfied:

$$\frac{Q_1}{Q_2} = \frac{N_1}{N_2} = \left(\frac{H_{T_1}}{H_{T_2}} \right)^{\frac{1}{2}} \quad (6.b)$$

where N_1 and N_2 represent different runner rotational speeds, Q_1 and Q_2 are the flow rates at speed N_1 and N_2 , respectively, and H_{T_1} and H_{T_2} is the head drop imposed by the turbine at speed N_1 and N_2 , respectively.

As a result, according to (3) and (6.b) the same H_T can be achieved with at least one α_1 and N combinations which means that turbine can impose the same head drop under different hydraulic efficiencies. Therefore, as Q changes with time t , we obtain the time-dependent lower and upper bounds $P_{T,\min}(t)$ and $P_{T,\max}(t)$ of turbine generation,

respectively:

$$P_{T,\min}(t) = \gamma \cdot Q \cdot H_T \cdot n_{h,\min} \quad (7.a)$$

$$P_{T,\max}(t) = \gamma \cdot Q \cdot H_T \cdot n_{h,\max} \quad (7.b)$$

where $n_{h,\min}$, $n_{h,\max}$ are the lower and upper bounds of the hydraulic efficiency (slack variable) that achieve $H_T = H_{PRV}$ under the adjustment of α_1 and N in a bounded range, respectively. These values are used as variables in the optimization process, as is was already presented in Bideris-Davos and Vovos (2023), and are then inserted to the OPF formulation as additional hydro-generator constraints. The details of this process are presented in Section 4, as part of the IPWDS model.

2.3. Relaxation of the turbine's generation limits -a trade-off between PDS and WDS

During specific time periods, the pressure regulating turbines in the WDS may produce power than cannot be absorbed directly from the PDS, resolving to a non-realistic solution if specific action is not taken. If the WDS operator can tolerate deviations from the ideal pressure set-points (e.g., the standards of the WDS consumers are still met), then the aforementioned turbine power limits ($P_{T,\min}, P_{T,\max}$) could be altered. Since turbines are strictly allowed to impose lower head drops than the initial PRVs ($H_T \leq H_{PRV}$), nodal pressure constraints could be relaxed towards a higher, but tolerable pressure for the consumer, i.e., the relaxation of $P_{T,\min}$ exclusively.

Therefore, in cases when the turbine power that can be absorbed by the PDS at time t , $P_T^f(t)$, is even lower than the current lower limit of turbine generation, i.e., $P_T^f(t) < P_{T,\min}(t)$, hydro-turbine is allowed to alter its H_T . If the excess pressure is not economically justified, then hydro-turbine is forced to operate at its lower limit ($P_{T,\min}$) and the excess power ($P_{T,\min}(t) - P_T^f(t)$), has to be dissipated to a dump or non-critical load (e.g. electric boiler) so that the net real power injection to the PDS is maintained at $P_{T,f}(t)$. As it evident, in this case hydro-turbine continues to operate as PRV equivalent with minimum PRD without altering its H_T . However, when it is economically justified, then turbine power generation limits are converted to soft limits in the OPF formulation at the expense of a higher PRD (H_T is altered) and $P_T = P_T^f$ can be directly injected to the PDS.

In the mathematical formulation of the problem, whenever such soft limits are active, a binary flag $f \in \{0, 1\}$ is raised ($f = 1$). New hard limits are determined according to the divergence from the ideal pressure standards. This value and the respective cost assigned to it, solely depend on how much is the turbine allowed to deviate from its specific and pre-determined pressure regulation requirements. The downstream node head before and after $P_{T,\min}$ relaxation, is calculated by (8.a), (8.b), respectively, while (8.c) ensures that the resulting head value ranges within the statutory limits set by (1).

$$H_j^{\text{initial}}(t) = H_j^{\text{start}}(t) - H_{T,\min}^{\text{initial}}(t) \quad (8.a)$$

$$H_j^{\text{relaxed}}(t) = H_j^{\text{start}}(t) - H_T^f(t) \quad (8.b)$$

$$MHS_{\min} \leq H_j^{\text{initial}}(t) \leq H_j^{\text{relaxed}}(t) \leq MHS_{\max} \quad (8.c)$$

where H_j^{start} is the total head at the start node of the pipe, H_j^{initial} and H_j^{relaxed} is the total head at the end node of the pipe before and after the $P_{T,\min}$ relaxation, respectively, and $H_{T,\min}^{\text{initial}}$ and H_T^f is the headloss imposed by the turbine before and after the $P_{T,\min}$ relaxation, respectively, while $H_{T,\min}^{\text{initial}}(t) \geq H_T^f(t)$.

The new hard limit P_T^f for turbine power generation after the $P_{T,\min}$ relaxation, is given by:

$$P_T^f(t) = \gamma \cdot Q \cdot H_T^f(t) \cdot n_{h,\min}, \text{ where } P_T^f(t) \leq P_{T,\min}(t) \quad (9)$$

As it is evident, cases when the turbine power production cannot be absorbed from the PDS, resolve to a solution where the individual optimization of one system competes with the other. More specifically, when the WDS constraints are binding, i.e., $P_T(t) = P_{T,\min}(t)$, WDS operates efficiently according to its individual requirements (minimum PRD), but the overall efficiency from PDS perspective is suboptimal (excess power must be dissipated). Conversely, when WDS constraints are flexible, i.e., $P_T(t) = P_T^f(t) \leq P_{T,\min}(t)$, power generation is optimized (WDS power production and injection to the PDS coincide) at the cost of WDS service level which diverts from the ideal condition for pressure regulation $H_T = H_{PRV}$. Obviously, under a co-optimization frame, the penalty cost of detaching from ideal pressure regulation must be determined by both PDS and WDS in order to ensure an overall optimal operation. A simple approach to express this cost without the

need of extensive information of the entire water network, is through the estimation of the additional water losses in the form of leakage created by the divergence from the ideal pressure point. In this work, the benefit from reduced pipeline water losses due to lower pressure is considered during the correlation of the trade-off between the violation of WDS ideal pressure setpoint and the feasibility of the OPF solution. For this reason, $P_{T,\min}$ should be allowed to relax only when the revenue collected by the turbine generation at $P_T(t) = P_T^f$ overcomes the cost of increased pipeline water losses during divergence from the ideal pressure setpoint.

These water losses are formulated as:

$$Q_{leak}(t) = \left(0.5 \cdot a_c \cdot \sum_{i=1}^{n_i} L_{ij} \right) H_j^{\beta_e}(t) \quad (10)$$

where Q_{leak} is the leakage flow at time t (l/s^{-1}), L_{ji} is the length of pipe i connected to node j , n_i is the number of pipes connected to node j , while

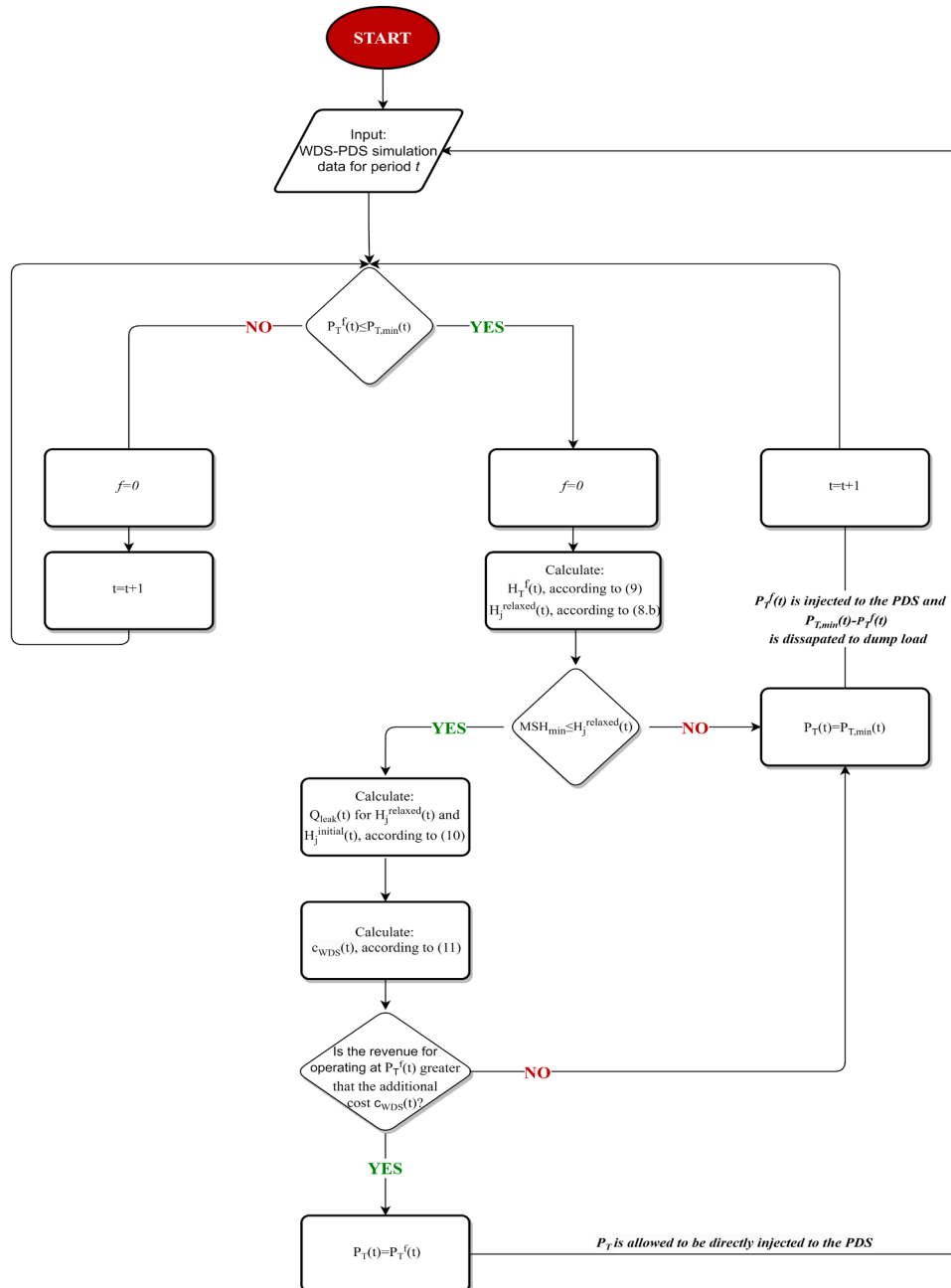


Fig. 1. Flow chart for the relaxation of the turbine's generation limit based on the trade-off cost c_{WDS} .

α_c and β_e are the leakage coefficient and exponent, respectively.

These losses are then penalized by the WDS operator with a cost c_{WDS} (\$/h), according to WDS's cost of water production c_w (\$/m³):

$$c_{WDS}(t) = c_w \cdot 3600 \cdot \frac{[Q_{leak}^{initial}(t) - Q_{leak}^f(t)]}{1000} (\$/h) \quad (11)$$

where $Q_{leak}^{initial}(t) - Q_{leak}^f(t) > 0$ represents the increase in pipe water losses due to the overpressure created at its end, caused by the sub-optimal pressure regulation of the turbine during hour t , as calculated by (10). The scale of this cost relies on the “distance” between $H_j^{initial}$ and $H_j^{relaxed}$ that is ultimately translated as the “distance” between $P_{T,min}(t)$ and $P_T^f(t)$, respectively. In order to implement this cost directly within the objective function of the co-optimization problem of Section 4, we incorporate it in a penalty function as:

$$c_{WDS} \cdot (P_{T,min}(t) - P_T^f(t)) (\$/kWh) \quad (12)$$

The flow chart that describes the process for the proposed generation limit relaxation is summarized in Fig. 1. Finally, it is worth mentioning that, if detailed data available, more complex expressions can be potentially adopted by the WDS operator to describe the cost of deviation from ideal quality of service that utilize extensive data from the entire WDS and account other factors, like consumer discomfort from low or high pressures.

3. Power distribution system model

In this Section, the PDS model and the integration of the hydro-turbine as a power generation device into the OPF formulation will be explained. OPF is a fundamental optimization tool that is used for determining the optimal power dispatch of an operating set of generators that satisfy the electric demand, while complying with various technical constraints. Here, it is extended in order to consider the impact of hydro-turbines on the PDS and create the link between the two systems.

PDSs differentiate from power transmission systems mainly due to their distinctive radial structure (zero or very small number of loops), their high line resistance to reactance ratio (R/X) and by the fact that there are no voltage controlled buses (negligible shunt capacitances and few or no generator buses). Thus, the simplified DC OPF is not applicable in PDSs, because the simplifying approximations used (e.g., $R = 0$) cannot be considered as valid. Typically, the inclusion of RES in the OPF problem, leads to OPF reformulations that occupy probabilistic models to deal with the uncertainty inserted by their intermittent output. Hydro-turbine generation in urban areas is a RES, but a priori well-bounded due to: a) the specific turbine head drop requirement (H_T) deriving from the PRV action and b) the well-predicted water flow pattern in WDS of urban areas (Bideris-Davos and Vovos, 2023). This enables the employment of the classical OPF formulation to be solved by deterministic methods, such as the one presented in (Wu et al., 2018). Therefore, hydro-turbines are henceforth considered as conventional generators with specific power generation limits depending on the water flow distribution in the WDS at a specific point in time.

Furthermore, most system operators are obliged to absorb as much renewable power as it is technically feasible and, in most countries, this is done under unity power factor (PF=1). In the OPF formulation these practices are simulated by attaching negative cost functions to the output of hydro-generators and setting their reactive power output to zero.

The general OPF formulation is as follows.

Objective function:

$$\langle \min \rangle f(x) \quad (13.a)$$

Subject to:

$$g(x) = 0 \quad (13.b)$$

$$h(x) \leq 0 \quad (13.c)$$

$$x_{\min} \leq x \leq x_{\max} \quad (13.d)$$

where $f(x)$ is the objective function, $g(x)$ is the set of non-linear equality constraints of real and reactive power balance equations, $h(x)$ is the set of non-linear inequality constraints (in our case just transmission constraints) and $x_{\min}$ and $x_{\max}$ are the feasible bounds of the optimization vector $x = [P_G, Q_G, V_m, \theta]^T$, where P_G, Q_G is the total power injected by a set of conventional and hydro-turbine generators, V_m is the bus voltage magnitude and θ the bus voltage angle.

The objective function typically consists of the costs of active and reactive generator injections:

$$\langle \min \rangle f(x) = \sum_{k=0}^{n_g} C_P^k P_G^k + C_Q^k Q_G^k \quad (13.e)$$

where n_g is the total number of generators and C_P, C_Q are the assigned active and reactive power costs of the generators. This formulation is altered in order to accommodate the hydro-turbine generation model as RES ($P_G = P_T, Q_G = Q_T = 0$) at each time period t as follows:

$$\langle \min \rangle \sum_{m=0}^{n_T(t)} -C_T P_T^m(t) \quad (13.f)$$

where $n_T(t)$ is the number of active turbines at period t , C_T is the cost coefficient assigned to the respective turbine real power output, $P_T(t)$. It is assumed that C_T represents the fixed feed-in tariff (FIT) offered by the energy market regulator for the hydroelectricity exported to the PDS side. In this way, the final payments will be representative of the actual monetary benefit of RES generation of a real case.

In (13.f) the factor -1 forces the operator to dispatch as much hydropower as technically feasible. Different negative cost factors among turbines could signify a preference of different FITs for hydro-generation at specific locations. More general objective functions could consider the output of other RES or the fuel cost of other conventional generators. In order to focus better on the link hydro-turbines create between WDS and PDS, the objective function is kept as simple as possible by considering only hydro-generation, with no preference over locations for power injection. The absence of conventional generators also relieves the problem from a possible unit-commitment problem that would rise from the multi-period time frame. A more detailed formulation that will include all aspects above will be the target of future work.

In power flow studies of transmission systems with voltage support at buses, constant P-Q load models are preferred. In this paper, WDS and PDS are supposed to overlap over an urban area, as it is the most common case. Thus all relevant assumptions for a distribution system are considered. Since distribution system loads are sensitive to voltage, a time-varying and voltage-dependent mixed load model is inserted in power flow equations (13.b) and expressed as follows (Hung et al., 2014):

$$\begin{aligned} P_{net,i}(V, t) &= P_{L,i}^{nom}(t) \cdot V_{m,i}^{\alpha_p}(t) \\ Q_{net,i}(V, t) &= Q_{L,i}^{nom}(t) \cdot V_{m,i}^{\beta_q}(t) \end{aligned} \quad (14)$$

where $P_{L,i}^{nom}(t)$ and $Q_{L,i}^{nom}(t)$ are the real and reactive power components of load at bus i , calculated at nominal voltage V_m^{nom} , $V_{m,i}(t)$ is the voltage magnitude at bus i at time period t and α_p, β_q are the active and reactive power exponents of $V_{m,i}(t)$. In the presence of extensive historical data, such power exponents can be specified in a PDS model, so that different consumer types with diverse load patterns and varying seasonal load behavior can be simulated accurately.

4. The interconnected power and water distribution system model

In (13.f), the optimization objective function has been set as the maximization of the total energy production of an operating set of turbines, specifically designed to replace existing PRVs over the course of a specified time frame. Turbine geometries are calculated so that they mimic PRV action with high precision, while their design is produced in such a way that their outer diameter always matches the corresponding pipe diameter. It has to be noted that a typical PRV installation does not include only the valve itself, but requires additional hydraulic equipment to ensure its functionality: upstream and downstream isolation valve, air valve and strainer etc. Therefore, a one-by-one replacement becomes technically feasible for WDS integration as the total PRV size becomes comparable with that of the designed turbine. As a result, the expected hydraulic conditions (nodal pressures and pipe flows) match the ones provided by the former WDS under PRV operation, and thus, the design process can be directly incorporated in the OPF formulation. The readers are encouraged to refer to previous work (Bideris-Davos and Vovos, 2023), for more information about the specifics of this process.

The standard OPF formulation is extended in order to accommodate turbines' power injection for each hour t , considering the respective performance bounds on the rotational speed N and inlet guide vane angle α_1 that satisfy the head loss requirements imposed by the former PRV action. Without loss of generality, as explained in Section 3, the co-optimization framework is simplified under the assumption that turbine generators are treated as RES by the PDS operator, providing only real power P_T (i.e., $Q_T = 0$) to the respective buses. The optimization process explained in Section 2, determines the turbine's operating point in order to concurrently maximize the real power injection at the PDS and impose a similar pressure drop as the previously installed PRV, according to the water flow. Since the power system operator is assumed to allocate hydro-generation as a RES, absorbing the maximum power that is technically feasible, the objective function is set to maximize the turbine's power output subject to the several hydraulic and power flow constraints that are discussed in Section 2 and Section 3.

The final problem formulation is expanded with additional variables and soft limits on the lower bound of turbine generation that reflect the ability of the WDS to operate on higher pressures than the initial PRV settings ((8)-(12)), at a measurable cost for service quality. A new optimization vector z is then formed that includes the OPF optimization vector x augmented with vector y that is imposed by the WDS-turbine model:

$$y = y(t) = [P_T, \alpha_1, N]^T \quad (15.a)$$

$$z = [x; y] \quad (15.b)$$

During instances when the turbine power production cannot be absorbed from the PDS, turbine is allowed to diverge from "PRV equivalent" operation. This divergence, translated in power generation, is penalized within the objective function with the incorporation of an additional cost that depends on the "distance" between $P_T^f(t)$ and $P_{T,\min}(t)$. The cost (c_{WDS}) associated with the penalty function is determined by both PDS and WDS, allowing a feasible P_T^f to be reached, only when it is proven to be cost-effective for the overall co-optimization. The unified optimization model formulation of the IPWDS is presented as follows.

Objective Function:

$$\langle \min \rangle \left[\sum_{k=1}^{j-1} C_P^k(P_G^k) + \sum_{k=j+1}^{n_g} C_P^k(P_G^k) + \sum_{m=1}^{n_T(t)} \left[(C_P^j(P_G^j) - C_T^m P_T^m(t)) + f \cdot c_{WDS} \cdot (P_{T,\min}^m(t) - P_T^f(t)) \right] \right] \quad (16.a)$$

Subject to:

$$F(\theta, V_m, P_{net}, Q_{net}) = 0 \quad (16.b)$$

$$|S(V_m, \theta)| \leq S_{max} \quad (16.c)$$

$$P_{G,\min} \leq P_G \leq P_{G,\max} \quad (16.d)$$

$$Q_{G,\min} \leq Q_G \leq Q_{G,\max}$$

$$V_{m,\min} \leq V_m \leq V_{m,\max}$$

$$\theta_{\min} \leq \theta \leq \theta_{\max}$$

$$\left[\begin{array}{l} P_{T,\min}(t) - f \cdot [P_{T,\min}(t) - P_T^f(t)] \leq P_T(t) \leq P_{T,\max}(t) \\ N_{\min}(t) \leq N(t) \leq N_{\max}(t) \\ a_{1,\min}(t) \leq a_1(t) \leq a_{1,\max}(t) \\ MHS_{\min} \leq H_j \leq MHS_{\max} \\ PRD \leq 2\% \\ H_T = \zeta_f \cdot Q^2 + \frac{u_2}{g} \left(\frac{1}{\pi D_2 b_1} \frac{1}{tg \alpha_1} + \frac{1}{\pi D_2 b_2} \frac{1}{tg \beta_2} \right) \cdot Q - \frac{1}{g} \left(u_2^2 + \frac{c_a}{2} + \frac{k \cdot w_c^2}{2} \right) \\ \frac{Q_1}{Q_2} = \frac{N_1}{N_2} = \left(\frac{H_{T1}}{H_{T2}} \right)^{\frac{1}{2}} \\ H_P = A Q^B \\ Q_{leak}(t) = \left(0.5 \cdot a_c \cdot \sum_{i=1}^{n_i} L_{ij} \right) H_j^{\beta_c}(t) \end{array} \right] \quad (16.e)$$

$$f = \begin{cases} 0, & P_{T,\min} \leq P_T^f \leq P_{T,\max} \\ 1, & P_T^f < P_{T,\min} \end{cases} \quad (16.f)$$

where (16.a) is the objective function of the IPWDS considering the trade-off cost between the violation of WDS pressure standards and the feasibility of the OPF solution of PDS, (16.b) are the AC Power Flow equations, (16.c) are the transmission constraints, (16.d) are the standard AC OPF constraints imposed by the PDS side, (16.e) are the constraints imposed by the turbine and WDS side and (16.f) is the binary flag for soft limit activation of pressure regulation. Equations (16.a)-(16.f) are calculated for each time period t of the optimization time frame H ($\forall t \in H$). During instances that feasibility of the OPF solution is guaranteed, i.e., $P_{T,\min} \leq P_T^f \leq P_{T,\max}$, the problem of (16.a) is simplified to a Non-Linear Problem (NLP) by setting $f = 0$, while during instances that infeasibility of the OPF solution is encountered, i.e., $P_T^f < P_{T,\min}$, it is converted to a MINLP by setting $f = 1$.

The cost assigned to the divergence from ideal pressure regulation is reflected through the cost $c_{WDS}(t)$ that essentially represents the trade-off between WDS and PDS operators in order to ensure feasibility in the OPF solution with the minimum deviation from WDS standards. Practically, it implies that when infeasibility in OPF solution is met, $P_T(t)$ will be co-optimized so that $P_{T,\min}$ is relaxed to P_T^f only when it is profitable for the interconnected system, i.e., when the revenue collected by the turbine operating at $P_T(t) = P_T^f$ overcomes the cost of diverging from the ideal pressure set point. Otherwise, pressure set point is considered as strict, pressure regulation offered by the turbine is ideal but the power surplus ($P_{T,\min} - P_T^f$) is dissipated to a non-critical load.

5. Simulation results

5.1. Example: IPWDS Description

The suggested co-optimization method has been tested on an interconnected PDS and WDS, both of reasonable sizes. For the WDS, a real scale system in Kentucky with typical infrastructure that presents both loop and branched configuration, is selected. It consists of one water

treatment plant, 28 tanks, 21 pumps and approximately 450 km of pipes whose diameter ranges from 40 to 400 mm. It has an average demand of 8774 m³/d, and supplies 5330 costumers (95% residential), while the total serviceable population counts up to 25320 with more than 11000 households. The total number of PRVs installed is 15, so according to (Ferrarese and Malavasi, 2020) an average of 150 kW of total power is dissipated from PRVs during each hour of the day. The water losses are estimated around 20% of the total water provision. The cost of water production is taken $c_w = 1\$/m^3$, that corresponds to an average cost for water utilities worldwide (Wittholz et al., 2008), while the pipeline leakage coefficient and exponent are assumed $\alpha_c = 10^{-4}$ and $\beta_e = 1.1$, with respect to the respective pipe material (PVC), according to (Ávila et al., 2021). The schematic overview of the IPWDS is depicted in Fig. 2. It includes the layout of the 69-bus PDS, the configuration of the WDS with indication on the location of the former PRVs, and the buses at which the hydro-turbines will be connected to create the link between the two systems. The PDS is derived from a portion of the PG&E system (Baran and Wu, 1989). The electrical demand of the 69-bus distribution system is assumed to follow a typical mixed load pattern on a summer day (Taleski and Rajcic, 1997) for three different load types: residential, commercial and industrial (Hung et al., 2014). The active and reactive power exponents for (14) are derived from Bibliography on Load Models for Power Flow and Dynamic Performance Simulation, 1995 and are

presented in Table 1. During each hour t , the load $P_{L,i}(t)$ and $Q_{L,i}(t)$ are deemed equal to the peak load provided by the base case, multiplied by the load factor that corresponds to each load type (Fig. 3). Typically, for micro hydropower connected to the grid, a FIT value of $C_T = 0.10\$/kWh$ is considered, as drawn from “Feed-In-Tariffs Policy on Wind, Biomass (2012). The water demand pattern of the WDS consumers in comparison with a base case demand is also displayed in Fig. 3. It can be observed that the peaks of electrical and water demand of the two systems largely coincide. PDS base case load sums up to 3.8 MW and WDS base case water demand sums up to an average of 304 m³/h. The reader is encouraged to refer to Baran and Wu (1989) and “Water Distribution System Operation (2024) for more information regarding the nature of the data used for the simulated IPWDS.

The turbine selection and dimensioning process for the replacement of PRVs has already been addressed in Bideris-Davos and Vovos (2023).

Table 1
Power exponents for different types of voltage-sensitive loads.

Load type	α_p	β_q
Residential	0.72	2.96
Commercial	1.25	3.5
Industrial	0.18	6

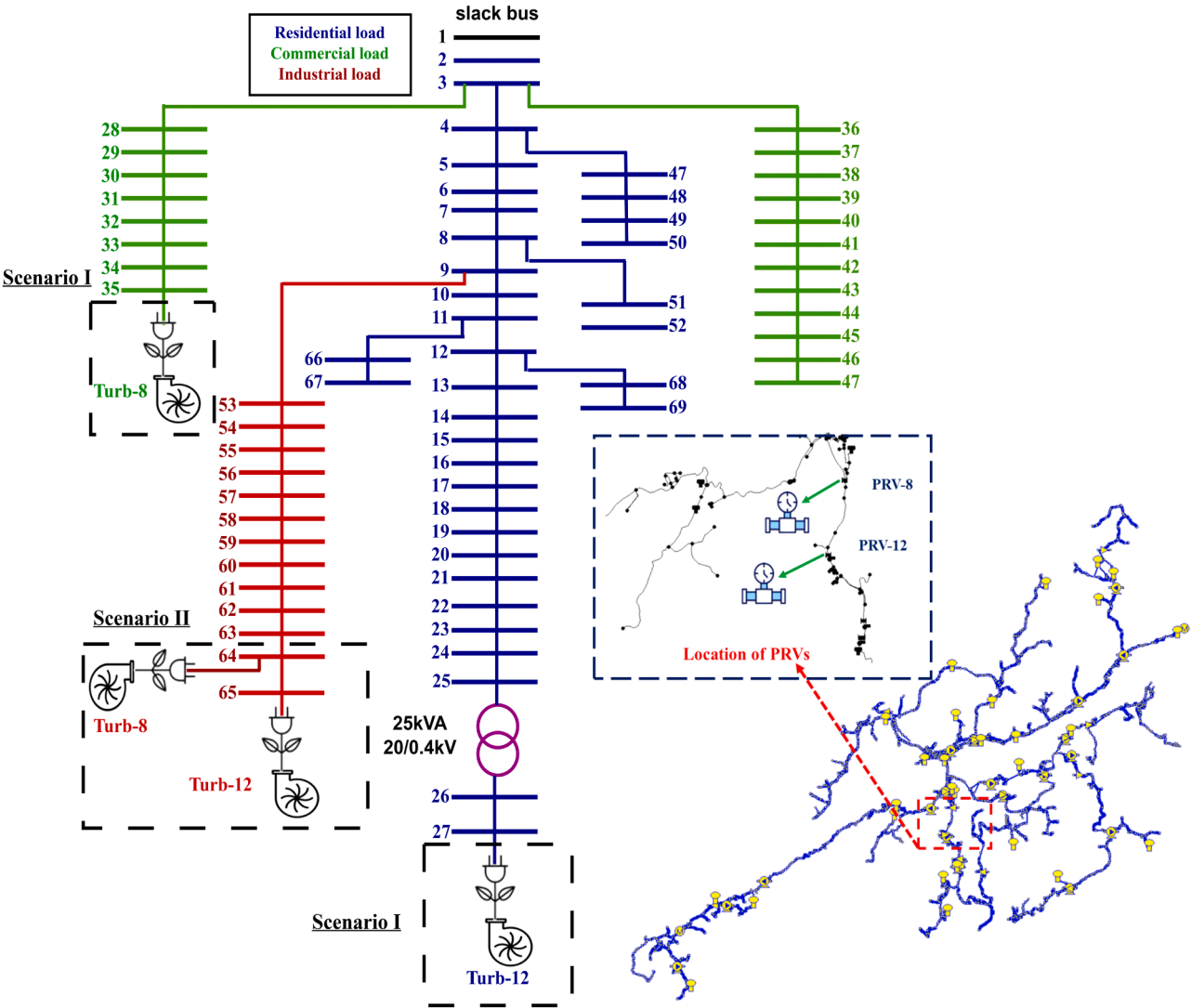


Fig. 2. Configuration of the IPWDS that uses pressure regulation turbines as intersystem links.

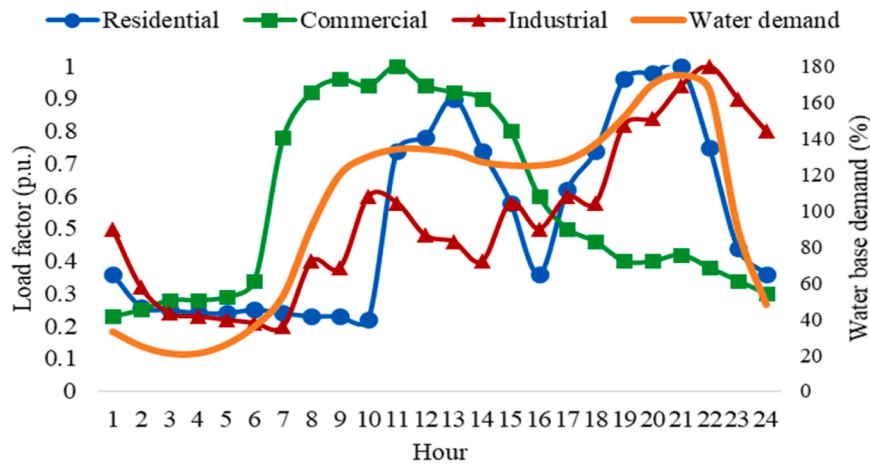


Fig. 3. Electrical loads and water demand profile of the IPWDS under study.

Two PRV locations from the WDS, namely PRV-8 and PRV-12, are replaced by Francis turbines (Turb-8 and Turb-12, henceforth) for the following reasons. Their individual installation can be economically justified due to the high energy yield that they present, leading to a real power production in the same order of magnitude (or even greater) as the PDS loads. They are also installed fairly close to each other and can potentially connect at different or the same bus, which will allow the creation of different scenarios of the same hydro-generator for two alternative locations of the PDS. All the important design and operational properties of the proposed hydro-turbines are presented in Table 2: design volumetric flow rate, design head, design rotational speed of the runner, specific speed, hydraulic efficiency at design point, number of blades, guide vane angle at design point and inlet and outlet diameter of the runner.

The operating bus voltage magnitude, is constrained between 0.95 p.u. and 1.05 p.u., as in most distribution systems (Baran and Wu, 1989; Bibliography on Load Models for Power Flow and Dynamic Performance Simulation, 1995; Hung et al., 2014). The MVA flow rating ($S_{\max}$) for each line is calculated according to standard ASCR overhead line conductors and considering each line length and conductor area (“Sustainable Energy Handbook, 2023). The connection to the transmission network is simulated by the slack bus of the simulated system (bus 1). For all cases, as it is explained in Section 3, turbine generators provide only real power P_T to the respective buses. Especially, for the case of rural PDS as the one presented in Fig. 2, a power injection with unity power factor ($PF = 1$), tends to impact bus voltages significantly (Vovos et al., 2007), which in turn impacts their power consumption, according to (14).

Table 2
Operational and design parameters of hydro-turbines.

Working Point Data	Turb-8	Turb-12
volumetric flow rate, Q (m ³ /h)	320	335
head, H (m)	38	27
rotational speed, N (rpm)	1800	1100
specific speed, n_q	35	28
hydraulic efficiency, η_h (%)	92	90
Design data		
Number of blades	6	6
guide vane opening angle, α_1 (°)	14	11
inlet diameter, D_1 (mm)	300	400
outlet diameter, D_2 (mm)	146	180
D_2/D_1 ratio	0.48	0.45

5.2. Base case

Based on the configuration of Fig. 2 and adopting the load curves of Fig. 3, power flows were carried out for a 24-h time frame ($H=\{1-24\}$) of PDS operation without the integration of hydro-turbines. The power system modeling and simulation is performed with the use of MATPOWER (Zimmerman et al., 2011), by using its default primal dual interior point method for solving NLPs, such as the one formulated in Equations (16.a)–(16.f).

A preparatory power flow analysis was executed to get a rough overview of the system condition and the voltage profile across the respective distribution lines and consumers. The p.u. voltage level of pilot buses during the day is displayed in Fig. 4. Slack bus voltage is set to 1.05 p.u.

Regarding the residential feeder line, it is observed that the high load is accumulated at its front-end and it gradually decreases after the middle of the line. Since the load level from bus 25 and henceforth is relatively low, a typical overhead 25 kVA transformer with 20/0.4 kV voltage rating, 2% no load losses and 4% impedance is assumed to supply the remaining households (Transformer Power, 2023). The transformer is also assumed to have reached its rated overloading level due to previous continuous loading and hence its thermal KVA limit is considered as a hard constraint during OPF. The transformer impedance is augmented with the impedance of the corresponding line that connects buses 25 and 26. Commercial load bus 35 exhibits voltages close to the upper permissible level due to its limited demand and its topological proximity to the slack bus. Industrial buses 64 and 65 exhibit the lowest voltage, due to the heavy demand conditions along the line (industrial loads) and their significant distance from the slack bus. During peak demand periods $t = \{20-23\}$, their voltage level is marginally above the lower permissible level of 0.95 p.u. The results of this analysis form the base case that will be used as a benchmark for PDS operation with turbines installed.

5.3. Co-optimization

The transformer location and the hydro-turbine integration within the PDS are illustrated in Fig. 2 and their impact is discussed in the following sub-sections. The IPWDS simulation is carried out for two different scenarios wherein the turbines are first connected at buses 27 and 35 (Scenario I) and finally at buses 64 and 65 (Scenario II). Each time period the binary flag is raised ($f = 1$), the penalty cost c_{WDS} is calculated according to (11) and (12) and the OPF is executed considering the trade-off between the violation of WDS ideal pressure setpoint and the feasibility of the OPF solution, see Sections 2.3 and 4. Computation time for all scenarios simulated on a computer with 1.8 GHz Intel

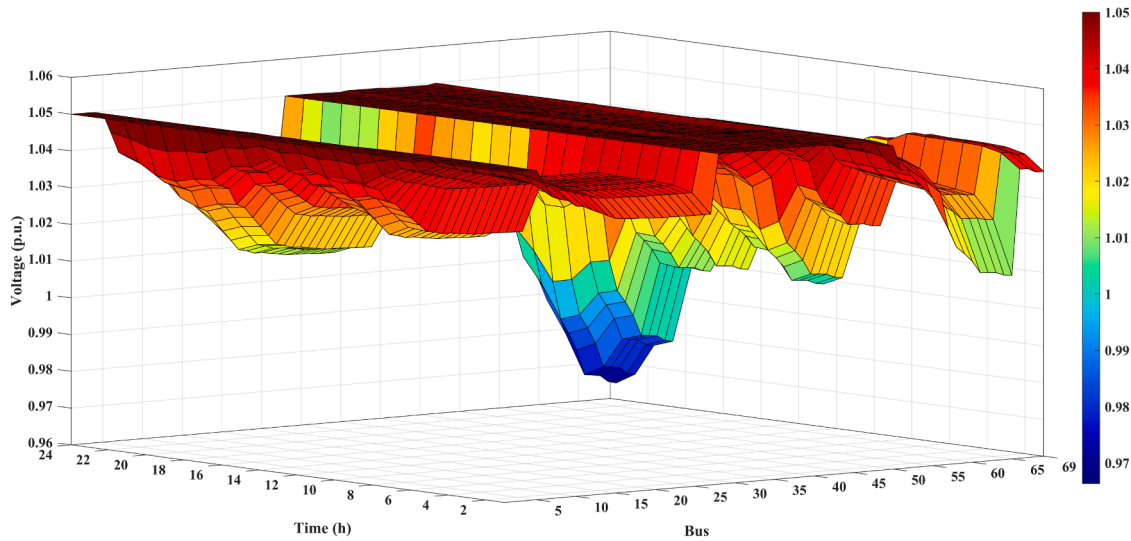


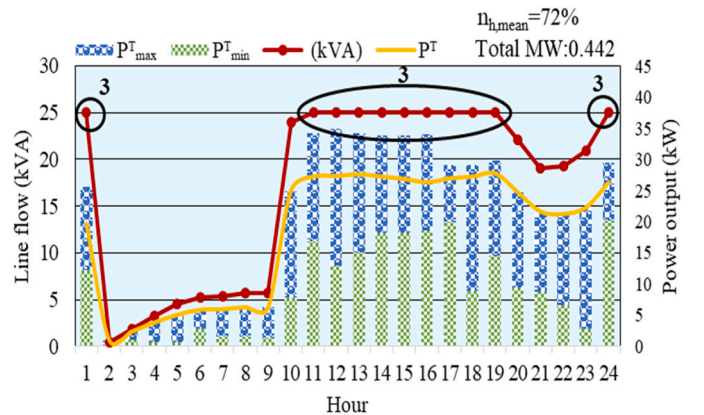
Fig. 4. Voltages of the PDS before the turbine integration.

i7 CPU/4 cores/8 GB RAM, has been less than 8 s, which can be considered negligible for planning.

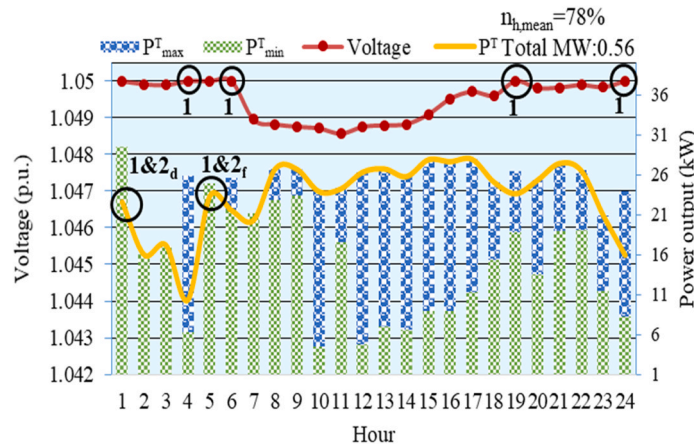
Scenario I

Under this scenario, Turb-8 and Turb-12 are connected at buses 35 and 27, respectively.

More specifically, Turb-12 is connected at terminal bus 27 of the “residential” line. In the base case, the power flow throughout the line is permanently unidirectional and the transformer thermal limit is always satisfied due to the limited load level of this area. Conversely, with the turbine integration, during periods of low local demand and high



a. Line flow through the transformer after Turb-12 integration at bus 27 (3: transformer thermal limit is reached)



b. Bus 35 voltage profile and power injection after Turb-8 integration (1: $V_{\max}$ is reached, 2: Soft limit is activated, subscript d: $P_{T,min}$ is inflexible and power is dissipated to dump load, subscript f: $P_{T,min}$ is flexible and no power is dissipated)

Fig. 5. PDS operation with respect to bus voltage and transformer limit constraints for Scenario I.

turbine output ($t = \{1, 24\}$) or periods of medium local demand and very high turbine output ($t = \{11-19\}$) the turbine power output surpasses the total load located beyond the transformer, the power flow is reversed and hits the KVA thermal limit of the transformer. Consequently, during $t = \{1, 11-19, 24\}$, the turbine output is restricted so that the transformer thermal limit is respected, as it is depicted in Fig. 5.a.

Furthermore, Turb-8 is connected at terminal bus 35 of the “commercial” line. For all time periods, except $t = \{1, 5\}$, the WDS pressure constraints are well expressed in the OPF formulation in the form of $P_{T,min}(t)$ and $P_{T,max}(t)$ and a solution that respects PDS and WDS constraints is reached, as it is depicted in Fig. 5.b. During time periods that is feasible (V_{max} is not reached at any bus and transformer loading limit is not reached), the OPF dispatch sets the power output of the hydro-turbine to its maximum, while in other periods, $t = \{1, 4, 5, 6, 19, 24\}$ the turbine power output is reduced because the maximum voltage V_{max} at bus 35 is reached. This confirms the general case for distribution systems that a power injection with unity power factor ($PF = 1$), tends to impact bus voltages significantly.

However, during periods $t = \{1, 5\}$, the real power generation of the Turb-8 exceeds the maximum permissible $P_T^f(t)$, that satisfies ideal WDS specifications. In other words, the turbine produces more power than the PDS can take, in order to regulate pressure at its ideal setpoint. According to Section 2, during these periods, turbine operation depends on the tightness of WDS pressure requirements. If the PRD cannot be altered, e.g., the WDS operator cannot reduce the minimum pressure requirement further, then turbine output is fixed to $P_{T,min}(t)$ but the power injection in the PDS remains fixed at $P_T^f(t)$. Solely in terms of simulation, this is done by setting $P_{T,min}(t) = 0$ at that location and accepting $P_{T,f}(t)$ received from the OPF as the power injected to the grid, while the excess power ($P_{T,min}(t) - P_T^f(t)$) is assumed to be dissipated at a dump or non-critical load. When the WDS pressure regulation constraints are considered flexible, the turbine attempts to reach $P_{T,f}(t)$ by relaxing $P_{T,min}(t)$. It is worth noticing that during $t = \{1, 2, 3, 5, 7\}$, P_T becomes binding as only one combination of α_1 and N was found that satisfies H_T under minimum PRD (i.e., $P_{T,min} = P_{T,max}$).

During $t = \{1, 5\}$ where $P_T^f(t) < P_{T,min}(t)$, binary flag f is raised and soft limits on $P_{T,min}$ are enabled, as displayed in Fig. 5.b. The total impact of pressure constraint relaxation on the allocation of Turb-8 power output considering the corresponding costs and subsequent PRD

increase, is displayed in Fig. 6, according to the process described in Fig. 1. For both periods, the turbine's head drop $H_j^f(t)$ that is required to reach $P_T^f(t)$, leads to the overpressures $H_j^{relaxed}$, according to (8.b), (9). The cost of the divergence from ideal pressure standards is then compared with the revenues from importing P_T^f in the grid, according to (11), (12). As it can be observed, during $t=1$, Turb-8 would need to deviate significantly from the ideal pressure setpoint (PRD would have to increase from 1% up to 24%) in order to reach P_T^f , hence leading to an economically prohibited solution for the IPWDS. At this instance, WDS constraints are considered as inflexible, $P_T = P_{T,min}$, and the power surplus $P_{T,min}(t) - P_T^f(t)$ is dissipated to a non-critical load, so that the net real power injection to the PDS is maintained at $P_{T,f}(t)$. Conversely, during $t=5$, the relaxation of $P_{T,min}$ is allowed ($P_T = P_T^f$), since the PRD increase from 0.8% to 6.7%, doesn't alter the pressure standards significantly and therefore this small relaxation remains cost-efficient for the IPWDS, as shown in Fig. 6.

Scenario II

Under this scenario Turb-8 and Turb-12 are assumed to be connected at adjacent buses 64 and 65, respectively. The significant voltage drop across the respective line, allows a great amount of localized hydro-power generation. For all time periods, since no voltage or line limit is reached, hydro-turbine power output is consumed locally and is fixed to its upper limit $P_{T,max}(t)$, as shown in Fig. 7. Again for Turb-8, as with Scenario I, P_T becomes binding during $t = \{1, 2, 3, 5, 7\}$, but $P_{T,min}$ is never relaxed since OPF feasibility is always guaranteed.

The results taken from the IPWDS using the suggested framework confirms that turbines are technically capable in regulating water pressure as efficiently as PRVs and exploit the substantial amount of dissipated hydropower that is available in large WDSs, while offering power generation with high efficiency. According to Laghari et al. (2013), the total installation cost of mini/micro hydro-turbine projects like the one proposed here, ranges between 1500 and 2500 \$ per kilowatt of installed capacity, while the total cost of PRVs based on the pipe diameter, is determined by García et al. (2019). The estimated results of this approximate financial analysis are presented in Table 3. The annual revenue created from the energy exports to the electric grid according to the energy tariff C_T , is roughly estimated around 204.5 (Scenario I) and 215.35 MWh (Scenario II) for Turb-8 and 161.3 (Scenario I) and 177.75 MWh (Scenario II) for Turb-12 and indicates that the upfront capital cost

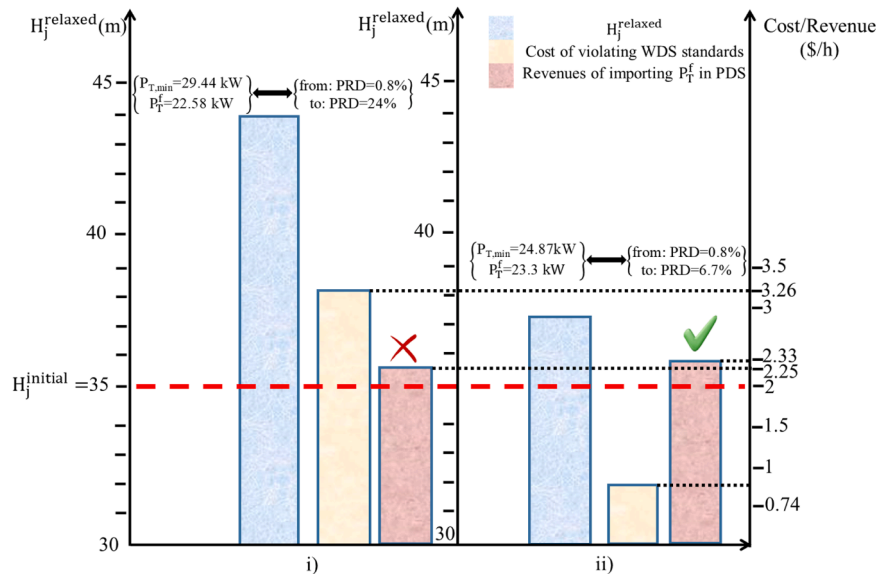
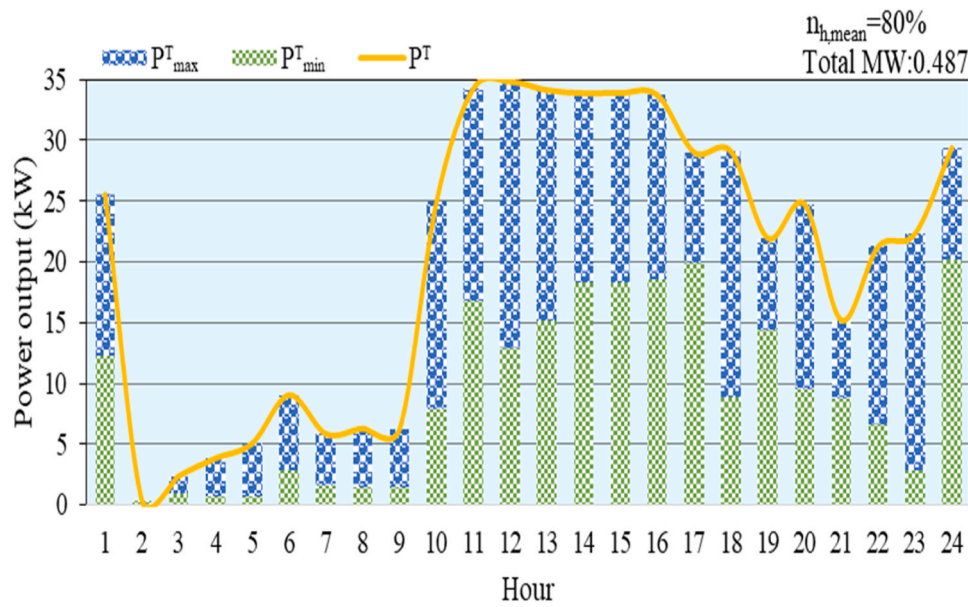
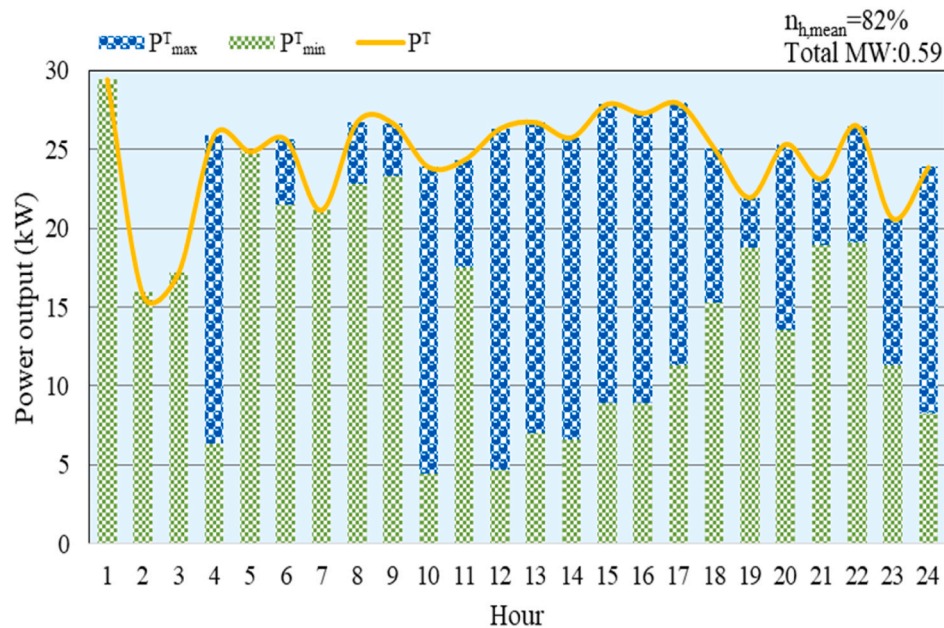


Fig. 6. Impact of pressure ($H_j^{initial}$) relaxation on the power allocated by Turb-8 during: i) $t=1$ which is not allowed and $P_T = P_{T,min}$. And ii) $t=5$ which is allowed and $P_T = P_T^f$.



a. Turb-12 power output at bus 65 (no line flow or voltage limit violations)



b. Turb-8 power output at bus 64 (no line flow or voltage limit violations)

Fig. 7. PDS operation for Scenario II.

of the turbines will be recovered in less than 4 years. These results are in accordance with the payback periods of micro hydroelectric projects, which typically range between 1 and 5 years (Laghari et al., 2013). However, realistically, PRVs should be replaced with turbines only when a detailed economic assessment has been carried out. Unfortunately, this requires a separate in-depth analysis that goes far beyond the scope of this work. It is the target of our future efforts.

PATs operating as energy recovery machines normally display an installation cost that is significantly lower compared to turbines. However, if along with energy recovery, they are required to operate as pressure control devices, their cost increases significantly, due to the additional equipment that is required in order to regulate pressure accurately, such as a strainer, a PRV and two flow control valves at their outlet. The average cost of PATs equipped with control and hydraulic components in order to offer pressure control is also displayed in

Table 3. This cost is lower, but comparable, with the one presented by turbines. On the contrary, their hydraulic efficiency is significantly inferior to the one presented by turbines. It typically ranges below 70%, as drawn from (Postacchini et al., 2020; Macías Ávila et al., 2021; Yang et al., 2012), while their payback period ranges from 2 to 10 years, as drawn from (García et al., 2019; Pugliese and Giugni, 2022). As it is evident from Figs. 5 and 7, the suggested turbine designs offer improved efficiency, high power generation, while their pressure regulation range is still superior than the one offered by PATs, without the need of external hydraulic equipment to achieve this. As it is shown in Table 3, their payback period when installed in sites with substantial energy recovery potential is within the expected range that is estimated for micro-hydropower units (between 2 and 5 years), according to (Laghari et al., 2013).

Table 3
Cost estimation and payback period of hydro-turbines.

Key inputs	Turb-8	Turb-12		
Installed capacity (kW)	33	24.6		
Investment cost of turbine (\$x10 ³)*	49.5–82.5	36.9–61.5		
Investment cost of PRV (\$x10 ³)**	19.5	33		
Investment cost of PAT (\$x10 ³ ***)	24	21		
Economic assessment	Scenario I	Scenario II	Scenario I	Scenario II
Annual revenue from energy exports (\$x10 ³)	20.45	21.53	16.13	17.77
Turbine payback period (yr)****	2.4–4	2.3–3.8	2.3–3.8	2.1–3.5

* According to (Laghari et al., 2013)

** According to (García et al., 2019)

*** According to (García et al., 2019; Novara et al., 2019)

**** Initial cost/Average annual net revenue

6. Conclusion

The replacement of PRVs with hydro-turbines creates links between the WDS and PDS. These links allow their co-optimization, improving the overall system efficiency by utilizing the otherwise dissipated hydropower, while providing water pressure regulation. The proposed method enables the incorporation of those turbines within the classic OPF formulation. Additional constraints that account for the turbine feasible operating region in terms of power production and pressure regulation were inserted. The objective function of the optimization problem was expanded to accommodate hydro-turbines as typical RES generators with specific lower and upper generation bounds. Divergence from the ideal pressure regulation can be accommodated, where this is feasible, but with an appropriate trade-off cost attached, that represents the compromise between WDS and PDS in order to ensure a profitable solution for the interconnected system. The unified co-optimization model was tested on real WDS and PDS of reasonable scale, over a 24-h time frame for two different scenarios. Under the first scenario, the turbines output respects PDS constraints and is restricted by a transformer thermal limit and the upper voltage limit. At the same time, the solution respects WDS pressure specifications under ideal PRD or sub-optimal PRD by allowing a relaxation on pressure regulation, on time periods where initial OPF is not feasible. During those instances, turbine's power allocation is determined according to the relative cost that such relaxation would create to both systems. Under the second scenario, since OPF does not violate PDS limit at any instance and feasibility is always guaranteed, both turbines' power output is maximized under minimum PRD. The results demonstrate that the WDS and turbine constraints are well-embedded in the OPF formulation. At all instances power generation is maximized, while respecting the PDS constraints, turbine's operational restrictions and the pressure regulation requirements imposed by the WDS. For all scenarios, the actual turbines' main dimensioning and working point were determined to precisely mimic the former PRVs operation, while producing power efficiently. The benefits, challenges and limitations of the suggested framework are summarized below.

The linkage between water and system infrastructure, could conceivably provide valuable services to both systems. The suggested framework can be used to model and simulate IPWDSs of different specifications and operational parameters (scale, flow rates, pressures, voltage levels etc.). The analysis of such a wide pool of results can be used for the proposition of general guidelines on how both WDS and PDS should expand in the future, so that a maximum penetration of hydro-turbines is secured. Via this unified model, water and power system operators can coordinate their actions and accomplish optimality of operation that is centrally monitored and updated. Different objective functions or constraints of weighted significance can be incorporated. For the PDS side, it can provide voltage stability at peak hours or minor contingencies, increase system spinning reserve capacity in the form of inertia response and provide cheap renewable energy during peak

demand. For the WDS side, the extensive energy requirements for the pumping and desalination systems could be partially supplied via the turbine operation, while achieving water savings by regulating pressure to lower, but sufficient level for consumers. Except the various practical challenges that are demonstrated in this paper during the integration of hydro-turbines on large scale WDSs and PDSs, there are still several technical, regulatory and social aspects that need to be considered, such as:

- **Technical:** PRV installation and operation does not include only the valve itself, but also requires additional hydraulic equipment to ensure its functionality, e.g., upstream and downstream isolation valve, air valve and strainer. Moreover, Francis turbines require a volute casing, where the guide vanes mechanism is hosted. Along with the turbine runner dimensioning that is presented in this paper, one of the challenges is to optimally determine the volute casing dimensioning as well, so that the total space occupied by the PRV installation can adequately accommodate the turbine installation, without the need of additional civil work. Another issue that needs to be addressed is how much more power are hydro-turbines allowed to produce by maintaining the minimum acceptable pressure to end-users and how can this consumer discomfort be quantified and incorporated in the trade-off cost.

- **Regulatory:** There are currently no licensing protocols for the new hydrogenation added in the urban environment. Who will own them and operate them (PDS operator, WDS operator or both)? How the profits will be divided between operators?

- **Social:** Local generation increases economic and energy independence of local communities. However, when integrating hydro-turbines into a weak PDS, the voltage rise that is created may have adverse impacts to operational reliability and consumers' service quality. More research needs to be done towards the optimal cooperation between voltage control mechanisms (centralized or distributed) in order to ensure a high penetration of hydroelectric generators within existing PDSs with minimized voltage impacts.

Future research directions should include the quantification of the shadow benefit from avoiding the purchase, operation and maintenance of a PRV that would perform the same job as a pressure regulating turbine, but without creating any monetary benefit from energy recovery. Finally, the existing bibliography, in order to determine the total cost of mini-micro turbines, currently employs cost functions derived from small hydro projects plants. However, this may lead to an over-estimation of the cost, since a significant part of the total cost of small hydro plants depends on civil work cost (50% on average). However, such a high percentage cannot be justified for hydro-turbines installed in WDSs. Unlike small hydro plants, micro-mini turbines in a WDS usually do not require the construction of dams or reservoirs. As a result, new, more accurate cost estimation models that are strictly derived and adapted to mini and micro turbine units should be developed. Finally, it is target of future work to investigate the impact of reactive power control of hydro-generators on both systems, as well as the implications the unit-commitment problem of conventional generators might have to

the combined operation of PDS and WDS.

Author Statement

We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed.

CRediT authorship contribution statement

Panagis N. Vovos: Writing – review & editing, Validation, Supervision, Formal analysis, Conceptualization. **Admitos A. Bideris-Davos:** Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare no conflict of interest.

Data availability

The authors are unable or have chosen not to specify which data has been used.

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